Ajith Kumar

Senior Backend Engineer | Node.js, TypeScript and AWS

kumarajith1996@gmail.com • kumaji.dev • github.com/kumarajith • linkedin.com/in/ajithk96

SUMMARY

Backend engineer with 8+ years of experience building Node.js and TypeScript services on AWS for a European vehicle marketplace. I enjoy owning systems end to end, from architecture and infrastructure to keeping them healthy in production. I like taking on new technical challenges and helping the engineers around me grow. Along the way, I've worked in Python and PHP, built an LLM product backend and led a small team.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL
Backend: Node.js, NestJS, FastAPI, Laravel
AWS: Lambda, Fargate, API Gateway, SQS, EventBridge, Step Functions, DynamoDB
Data: PostgreSQL, MySQL, DynamoDB, Redis, Kinesis
DevOps: AWS CDK, GitHub Actions, Docker

EXPERIENCE

ACV Auctions (now part of Copart) | Drive, its European car marketplace, ~50k vehicles a year — Chennai, India (Hybrid)
Senior Fullstack Engineer (SDE-4) — Jun 2026 – Present
Fullstack Engineer (SDE-3) — Jun 2025 – Jun 2026

- Designed the cloud architecture for replatforming a core application that sends ~30,000 vehicles to auction annually; provisioned infrastructure using AWS CDK and implemented approval-gated deployment pipelines with GitHub Actions.
- Owned the zero-downtime migration of a vehicle-history API used by five internal services from Heroku to AWS Fargate, upgrading Node.js from 18 to 24, adding CloudWatch and Sentry observability, and decommissioning Heroku.
- Designed and built a reusable data-access layer that unifies 5+ PostgreSQL, MySQL and other application databases behind a single API, so new services don't need their own connectors or credentials.
- Modernized a vehicle condition-report service generating ~1,000 PDFs weekly, migrating it from Google Cloud Functions to AWS Lambda, API Gateway and SQS; upgraded dependencies and added a dead-letter queue for failed jobs.
- Designed and built a unified transport-booking API supporting three provider integration patterns: REST APIs, CSV uploads and email parsing.
- Shared cross-region on-call with Ireland-based teams, writing runbooks for recurring production issues and working with PMs to turn repeat support problems into product features.
- Mentored a frontend-focused engineer through backend development and AWS implementation, reviewed teammates' system designs, and served as a secondary interviewer for senior and staff engineering candidates.

Tech: Node.js, TypeScript, NestJS, AWS (Lambda, API Gateway, Fargate, SQS, EventBridge, Step Functions, DynamoDB, CloudWatch), AWS CDK, GitHub Actions, Sentry, Jest, SQL

ZoomRx | B2B healthcare analytics SaaS — Chennai, India (Hybrid)

Senior Software Development Engineer II — Oct 2022 – Apr 2025

Senior Software Development Engineer I — Apr 2021 – Sep 2022

Synapse is ZoomRx's survey analytics platform, built on PHP/Laravel, MySQL and React and used by 100+ people running 1,000+ queries a day.

- Reduced average query latency by 50% on Synapse through schema changes, join restructuring and composite indexes, validated by replaying two years of real user queries on matched old and new instances.
- Took ownership of a Python data warehouse spanning 1,000+ market-research studies and millions of data points, extending it to power client-facing Tableau dashboards and reduce ad hoc reporting requests.
- Built the Python/FastAPI backend for a pharmaceutical market-research LLM platform, implementing structured intent extraction and multi-stage prompt orchestration to guide RAG retrieval and stream responses over SSE.
- Built the first release of a PowerPoint add-in that pulls Synapse data directly into presentations, replacing hours of manual Excel-to-slide work; it became the primary tool for ~90% of analyst decks.
- Built a custom Stripe billing layer for Ferma Chat supporting base plans, add-ons that stack on any plan, monthly and yearly terms, and prorated plan changes, after evaluating subscription providers.
- Led a two-engineer team for two years, running 1:1s and sprint planning, guiding technical design and release stability while remaining hands-on with backend development.

Software Development Engineer — Jun 2018 – Mar 2021

Built features for survey and admin portals, fixed production issues and supported high-priority customer incidents.

CERTIFICATIONS & PUBLICATIONS

AWS Certified Solutions Architect – Professional (SAP-C02) — Nov 2026

A Graph-Based Frequent Sequence Mining Approach to Text Compression — MIKE 2017

Published research conducted during an internship at IIITDM Kancheepuram.

SELECTED PROJECTS

Spotter | Serverless fitness-tracking Discord bot — 2026

- Built an event-driven Discord bot for daily fitness tracking, streaks and leaderboards, using Lambda and API Gateway for interactions, EventBridge Scheduler for daily jobs and DynamoDB single-table design. Has run for 6+ months at $0 monthly infrastructure cost.

Tech: Node.js, AWS Lambda, API Gateway, EventBridge Scheduler, DynamoDB, AWS CDK, Discord API

Watchparty Bot | Browser extension and companion app — 2026

- Built a Node.js companion server and Chrome extension for a Discord-controlled watch party, supporting synchronized play, pause and seek across Netflix, Prime Video and VLC. Isolated platform-specific integrations into modular provider components.

Tech: Node.js, Chrome Extension (MV3), Discord API, VLC API

EDUCATION

Bachelor of Engineering, Computer Science — 2014 – 2018

Sona College of Technology

CGPA: 8.97/10 | First Class with Distinction